Assignment - 02

Bachelor of Computer Application
Semester – V

Sub: Generative AI

Topic: Exploring Text Generation Using GPT-2
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Assignment 2: Exploring Text Generation Using GPT-2

Objective

This assignment looks at how changing different settings affects the text produced by GPT-2, a pretrained language model from the Hugging Face Transformers library.

Method

One opening sentence, "The scientist opened the door and saw", was used for five experiments. In each one, only one setting was changed at a time: temperature, top-k, top-p, or repetition penalty. After that, a custom prompt was used to create three more pieces of text using different settings.

Experiment 1: Temperature = 0.7

Output: "The scientist opened the door and saw a black box. He looked at it and stared at the screen, as though it had been a piece of paper with the words 'Science 101' painted on it. 'What the hell is this?'"

This text was calm and made sense from start to end.

Experiment 2: Top-k = 10

Output: "The scientist opened the door and saw a woman in a dress with a blue headband, a white t-shirt, and black pants with matching black shoes. 'Are you okay?' he said. 'No, I am fine...'"

This text stayed safe and simple, without much variety.

Experiment 3: Top-p = 0.9

Output: "The scientist opened the door and saw a man in a black leather chair sitting on the floor, watching. 'What is your name?' he asked. 'David. He's the guy who worked in a lab on a piece of...'"

This text felt natural and had a good mix of being sensible and interesting.

Experiment 4: Repetition Penalty = 1.5

Output: "The scientist opened the door and saw a woman, probably from other species. She looked at him with an expression she could only understand well: 'I am your father.' With that realization Lutz realized who he was talking about now-"

This text avoided repeating words and still made sense most of the way through.

Experiment 5: Temperature = 1.8 (High)

Output: "The scientist opened the door and saw someone in a dress looking over for Blake. 'Here's my friend that just dropped an important manuscript...' Weiss was sitting next to Blake. Weiss had just returned as seen the last message on the group's..."

This text jumped between ideas more and felt less predictable than the others.

What I Noticed

When the settings were low (like low temperature or low top-k), the text stayed safe, calm, and a little repetitive. When the settings were high (like high temperature), the text became more creative but also more random and harder to follow. Top-p and repetition penalty gave a good balance between the two, keeping the text interesting but still readable.

My Own Prompt

Prompt used: "Dear team, I am writing to inform you that"

Version 1 (temperature = 0.7):

"Dear team, I am writing to inform you that we have had the opportunity to discuss the recent decision to launch the National Broadband Network (NBN) in the UK. While we have... we have been very clear that we would like to make sure that each..."

Version 2 (top-k = 50):

"Dear team, I am writing to inform you that your team member has suffered an injury at the hands of Captain James Roberts of the U-3X, and we are conducting an examination of your actions. Thank you for your consideration."

Version 3 (top-p = 0.85):

"Dear team, I am writing to inform you that the following is an excerpt from an upcoming interview that I did with a colleague on Twitter: What I want to tell you is that, while I'm glad that you were able to stay with..."

Version 1 stayed the closest to a normal email. Version 2 turned into a strange story about an injury, which was unexpected. Version 3 became creative and mentioned an interview, showing how the same starting sentence can lead to very different results.

Bonus: Strange Output at High Temperature

At a very high temperature (2.0), the text became harder to follow and jumped between unrelated ideas. The output was:

"The scientist opened the door and saw another dark figure staring in a slightly darker shade of blue. 'He's probably drunk?' Professor Flicker asked and she froze in surprise—but even those few hundred and twenty dollars from his account could pay off in..."

This happens because a high temperature makes the model treat likely and unlikely words as almost equally probable. So it starts picking odd or unrelated words more often, which causes the sudden topic changes seen above.

Conclusion

Overall, this assignment showed that small changes in settings can have a big effect on what GPT-2 writes. Lower settings keep the text safe and predictable, while higher settings make it more creative but also more likely to lose sense. Top-p and repetition penalty were the most balanced choices out of all the ones tested.

Assignment 2: Exploring Text Generation Using GPT-2

Objective

This notebook explores how different text-generation parameters (temperature, top-k, top-p, and repetition penalty) affect the output of the pretrained GPT-2 language model from Hugging Face Transformers.

Approach

One opening sentence was used to run five experiments, changing one parameter at a time. A custom prompt was then used to generate three additional versions with different settings. The results are compared in a table, followed by an analysis and a bonus example of unusual output at high temperature.

```
# Step 1: Install required libraries (Hugging Face Transformers +
PyTorch)

!pip install transformers torch
```

```
Requirement already satisfied: transformers in
/usr/local/lib/python3.13/dist-packages (5.16.1)
Requirement already satisfied: torch in
/usr/local/lib/python3.13/dist-packages (2.11.0+cpu)
Requirement already satisfied: huggingface-hub<2.0,>=1.5.0 in
/usr/local/lib/python3.13/dist-packages (from transformers) (1.28.0)
Requirement already satisfied: numpy>=1.17 in
/usr/local/lib/python3.13/dist-packages (from transformers) (2.1.3)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.13/dist-packages (from transformers) (26.3)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.13/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex>=2025.10.22 in
/usr/local/lib/python3.13/dist-packages (from transformers)
(2025.11.3)
Requirement already satisfied: tokenizers<0.24.0,>=0.23.1 in
/usr/local/lib/python3.13/dist-packages (from transformers) (0.23.1)
Requirement already satisfied: typer in
/usr/local/lib/python3.13/dist-packages (from transformers) (0.27.1)
Requirement already satisfied: safetensors>=0.8.0 in
/usr/local/lib/python3.13/dist-packages (from transformers) (0.8.0)
Requirement already satisfied: tqdm>=4.60 in
/usr/local/lib/python3.13/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: filelock in
```

```
/usr/local/lib/python3.13/dist-packages (from torch) (3.32.4)
Requirement already satisfied: typing-extensions>=4.10.0 in
/usr/local/lib/python3.13/dist-packages (from torch) (4.16.0)
Requirement already satisfied: setuptools<82 in
/usr/local/lib/python3.13/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in
/usr/local/lib/python3.13/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/usr/local/lib/python3.13/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.13/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in
/usr/local/lib/python3.13/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: click<9.0.0,>=8.4.2 in
/usr/local/lib/python3.13/dist-packages (from huggingface-
hub<2.0,>=1.5.0->transformers) (8.5.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.5.2 in
/usr/local/lib/python3.13/dist-packages (from huggingface-
hub<2.0,>=1.5.0->transformers) (1.6.0)
Requirement already satisfied: httpx<1,>=0.23.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface-
hub<2.0,>=1.5.0->transformers) (0.28.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.13/dist-packages (from sympy>=1.13.3->torch)
(1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.13/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: shellingham>=1.3.0 in
/usr/local/lib/python3.13/dist-packages (from typer->transformers)
(1.5.4)
Requirement already satisfied: rich>=13.8.0 in
/usr/local/lib/python3.13/dist-packages (from typer->transformers)
(13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.13/dist-packages (from typer->transformers)
(0.0.5)
Requirement already satisfied: anyio in
/usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<2.0,>=1.5.0->transformers) (4.14.2)
Requirement already satisfied: certifi in
/usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<2.0,>=1.5.0->transformers) (2026.7.22)
Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<2.0,>=1.5.0->transformers) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.13/dist-
packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0-
>transformers) (3.19)
Requirement already satisfied: h11>=0.16 in
```

```
/usr/local/lib/python3.13/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (0.16.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.13/dist-packages (from rich>=13.8.0->typer->transformers) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.13/dist-packages (from rich>=13.8.0->typer->transformers) (2.21.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.13/dist-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer->transformers) (0.1.2)
```

```
# Step 2: Load the pretrained GPT-2 model and tokenizer

from transformers import GPT2LMHeadModel, GPT2Tokenizer

tokenizer = GPT2Tokenizer.from_pretrained("gpt2")
model = GPT2LMHeadModel.from_pretrained("gpt2")
```

{"model_id":"b5f87de5b1a246b7bcb97227aab0b8dc","version_major":2,"version_minor":0}

```
# Step 3: Choose one opening sentence to use for all 5 experiments

prompt = "The scientist opened the door and saw"

# Step 4: Prepare the prompt for the model and set up a reusable text-generation function

input_ids = tokenizer.encode(prompt, return_tensors="pt")

def generate_text(**kwargs):
    output = model.generate(
        input_ids,
        max_length=50,
        pad_token_id=tokenizer.eos_token_id,
        **kwargs
    )
    return tokenizer.decode(output[0], skip_special_tokens=True)

# Experiment 1: Change temperature (controls randomness of word choice)

result1 = generate_text(do_sample=True, temperature=0.7)
print(result1)
```

```
The scientist opened the door and saw a man in a black coat walking up the steps.

"Hey, that's me," he said.
```

```
"What's your name?" I asked.

"Me"

"I

# Experiment 2: Change top-k (limits word choice to the k most likely
next words)

result2 = generate_text(do_sample=True, top_k=10)
print(result2)

The scientist opened the door and saw the man standing at his window.
A short, bearded man with long blond hair, he walked up the stairs,
his face contorted with fear.

The man looked like he had been in his mid-

# Experiment 3: Change top-p (nucleus sampling – picks from smallest
set of words covering probability p)

result3 = generate_text(do_sample=True, top_p=0.9)
print(result3)

The scientist opened the door and saw an old man on a white dress
walking out in front of a window.

"Are you coming out here to ask him about me?" asked the man who was
sitting next to him. "I've been

# Experiment 4: Change repetition penalty (discourages repeating the
same words)

result4 = generate_text(do_sample=True, repetition_penalty=1.5)
print(result4)

The scientist opened the door and saw a black cloud rising. The
temperature was on high, but not yet over 400 degrees Fahrenheit (14
million F), at that point she thought it may be time to head back
outside for lunch or bedtime! But

# Experiment 5: Change temperature again, using a high value to see
more random/creative output

result5 = generate_text(do_sample=True, temperature=1.8)
print(result5)

The scientist opened the door and saw five children, two white girls,
nine-five, with blond faces on a blue mat wearing shorts, tennis shoes
on their back, orange baseball hats, black shorts—a black sports
apparel store owner came in and
```

```
# Step 6: Create a custom prompt of my own, to generate 3 different
versions

my_prompt = "Dear team, I am writing to inform you that"
my_input_ids = tokenizer.encode(my_prompt, return_tensors="pt")

# Custom Version 1: Generate using temperature = 0.7

output1 = model.generate(my_input_ids, max_length=50, do_sample=True,
temperature=0.7, pad_token_id=tokenizer.eos_token_id)
print(tokenizer.decode(output1[0], skip_special_tokens=True))

Dear team, I am writing to inform you that the following is a new
post. I was inspired to write this post because of my recent work with
the Google Docs team. This post will be updated as new information is
released.

We

# Custom Version 2: Generate using top_k = 50

output2 = model.generate(my_input_ids, max_length=50, do_sample=True,
top_k=50, pad_token_id=tokenizer.eos_token_id)
print(tokenizer.decode(output2[0], skip_special_tokens=True))

Dear team, I am writing to inform you that the team behind the latest
version of C++ is investigating the issues that might occur with the
current C++ implementation. A lot of people are using this in a very
simplistic way due to several reasons:

# Custom Version 3: Generate using top_p = 0.85

output3 = model.generate(my_input_ids, max_length=50, do_sample=True,
top_p=0.85, pad_token_id=tokenizer.eos_token_id)
print(tokenizer.decode(output3[0], skip_special_tokens=True))

Dear team, I am writing to inform you that my son and I are not going
to the White House for any of the press conference scheduled for
August 23. I am sure you understand that I am sorry that my son and I
would have to go
```

Experiment	Parameter Changed	Value	Output (short)
1	Temperature	0.7	"...saw a black box. He looked at it and stared at the screen... 'What the hell is this?'"
2	Top-k	10	"...saw a woman in a dress with a blue headband... 'Are you

Experiment	Parameter Changed	Value	Output (short)
			okay?'... 'No, I am fine...'"
3	Top-p	0.9	"...saw a man in a black leather chair... 'What is your name?'... 'David. He's the guy who worked in a lab...'"
4	Repetition penalty	1.5	"...saw a woman, probably from other species... 'I am your father.' With that realization Lutz realized..."
5	Temperature (high)	1.8	"...saw someone in a dress looking over for Blake... Weiss was sitting next to Blake..."

Analysis:

- Low temperature (0.7) produced fairly logical and predictable text, staying close to a normal story.
- Low top-k (10) limited the model's word choices, giving safe but slightly repetitive-feeling text.
- Top-p (0.9) gave a good balance of coherence and variety, since it allows a flexible pool of likely words.
- A higher repetition penalty (1.5) reduced repeated words and phrases, making the text a bit more varied but still readable.
- High temperature (1.8) produced more random and less predictable text, with sudden topic jumps and less logical flow.

Overall, lower parameter values (like low temperature or low top-k) made the text safer and more repetitive, while higher values (like high temperature) made it more creative and sometimes incoherent. Top-p and repetition penalty helped balance creativity and readability better than extreme temperature settings.

```
# Bonus: Generate using a very high temperature (2.0) to observe
unusual/incoherent output

result_bonus = generate_text(do_sample=True, temperature=2.0)
print(result_bonus)
```

```
The scientist opened the door and saw his wife's legs sway as she
passed and tried to pick her up from behind it in anger. So angry at
```

```
their son on some level, how had their kid been able to survive on
these people with only minor
```

Bonus - Unusual Output at High Temperature:

At temperature = 2.0, the output became less coherent. It started with "a dark figure in a slightly darker shade of blue," then unexpectedly shifted to a comment about the character being drunk, and then suddenly jumped to talking about "a few hundred and twenty dollars from his account."

This happens because a very high temperature flattens the probability differences between likely and unlikely words, so the model starts picking less probable (and sometimes irrelevant) words much more often. This causes sudden topic changes and less logical flow, since the model is no longer sticking closely to the most sensible next word.

```python
# Bonus: Generate using a very high temperature (2.0) to observe unusual/incoherent output

result_bonus = generate_text(do_sample=True, temperature=2.0)
print(result_bonus)
```

Bonus - Unusual Output at High Temperature:

At temperature = 2.0, the output became less coherent. It started with "a dark figure in a slightly darker shade of blue," then unexpectedly shifted to a comment about the character being drunk, and then suddenly jumped to talking about "a few hundred and twenty dollars from his account."

This happens because a very high temperature flattens the probability differences between likely and unlikely words, so the model starts picking less probable (and sometimes irrelevant) words much more often. This causes sudden topic changes and less logical flow, since the model is no longer sticking closely to the most sensible next word.

Custom Prompt Generations:

Prompt used: "Dear team, I am writing to inform you that"

Version 1 (temperature = 0.7): "Dear team, I am writing to inform you that we have had the opportunity to discuss the recent decision to launch the National Broadband Network (NBN) in the UK. While we have... we have been very clear that we would like to make sure that each..."

Version 2 (top-k = 50): "Dear team, I am writing to inform you that your team member has suffered an injury at the hands of Captain James Roberts of the U-3X, and we are conducting an examination of your actions. Thank you for your consideration."

Version 3 (top-p = 0.85): "Dear team, I am writing to inform you that the following is an excerpt from an upcoming interview that I did with a colleague on Twitter: What I want to tell you is that, while I'm glad that you were able to stay with..."

Observation: Version 1 stayed closest to a normal, sensible email. Version 2 introduced a strange and unrelated story about an injury, showing more randomness. Version 3 took a creative turn by referencing an "interview," showing that different sampling settings can push the same prompt into very different directions.